CNNs versus Vision Transformers for Dermoscopic Skin-Lesion Classification: A Controlled Comparison of ResNet-50, EfficientNet-B0 and Swin-T on ISIC 2019

José Alberto Martínez Navarro
Master's programme, Deep Learning course
Universidad Panamericana (UPE)
Aguascalientes, Mexico
jamn2000080300@gmail.com

***Abstract*—Automated classification of dermoscopic images is a mature application of deep learning, yet practitioners still face a practical question: should a clinical prototype be built on a convolutional backbone or on a vision transformer, and at what computational price? We present a controlled comparison of three ImageNet-pretrained architectures—ResNet-50, EfficientNet-B0 and Swin-T—fine-tuned on the eight-class ISIC 2019 dataset (25,331 images) under a strictly identical protocol: the same lesion-grouped stratified split, input resolution, augmentation, optimiser, schedule, class-weighted loss and early-stopping rule. Beyond accuracy and macro-F1 we report balanced accuracy, macro AUC, per-class F1, parameter count, FLOPs, peak training memory, training time, epochs to convergence and inference latency on a single consumer GPU, and we inspect the decision evidence of every model with Grad-CAM. On the held-out test split the best single model, Swin-T, reaches 64.1% macro-F1, while a soft-voting ensemble of the three backbones reaches 66.5% macro-F1 and 0.942 macro AUC. The results show that the accuracy gap between architecture families is small compared with their gap in cost, and that the choice should be driven by the deployment budget rather than by the family label.**

***Index Terms*—skin lesion classification, dermoscopy, ISIC 2019, convolutional neural networks, vision transformers, transfer learning, Grad-CAM, computational cost**

I. Introduction

Skin cancer is among the most frequently diagnosed cancers worldwide and melanoma, its deadliest form, is highly curable when detected early. Dermoscopy improves the diagnostic accuracy of trained dermatologists, but access to specialists is uneven, which motivates computer-aided diagnosis. Since Esteva *et al.* [1] showed dermatologist-level performance with a convolutional neural network (CNN), deep learning has become the standard approach, and the International Skin Imaging Collaboration (ISIC) challenges [2]–[4] have provided the public benchmarks on which progress is measured.

Two developments make a fresh comparison worthwhile. First, vision transformers (ViT, Swin) [5], [6] now rival CNNs on natural images, and their suitability for medical imaging, where datasets are small and class-imbalanced, is still debated [7]. Second, in clinical and embedded settings the relevant question is rarely "which model is most accurate" but "which model is accurate *enough* within a given latency, memory and training budget". Published results on ISIC are difficult to compare because they differ in splits, resolutions, augmentation, ensembling and metadata usage [8].

This work therefore asks a narrow, controlled question: *holding everything else fixed, how do a classical CNN (ResNet-50), an efficiency-oriented CNN (EfficientNet-B0) and a hierarchical vision transformer (Swin-T) compare on ISIC 2019 in accuracy, generalisation across classes, computational cost and interpretability?* Our contributions are:

- a reproducible, leakage-free experimental protocol (lesion-grouped stratified split, identical fine-tuning recipe) with open code;
- a multi-dimensional comparison covering accuracy, balanced accuracy, macro-F1, AUC, per-class behaviour, parameters, FLOPs, VRAM, training time, convergence speed and inference latency;
- a qualitative Grad-CAM analysis of what each architecture attends to, and a simple soft-voting ensemble that tests whether the families are complementary.

II. Related Work

Datasets. ISIC 2019 merges HAM10000 [3], BCN20000 [4] and the MSK dataset into 25,331 dermoscopic images labelled with eight diagnoses: melanoma (MEL), melanocytic nevus (NV), basal cell carcinoma (BCC), actinic keratosis (AK), benign keratosis (BKL), dermatofibroma (DF), vascular lesion (VASC) and squamous cell carcinoma (SCC). The distribution is strongly imbalanced (NV alone is about half of the data) and several images may depict the same lesion, which can inflate scores if the split ignores lesion identity [8].

CNNs on ISIC. The ISIC 2019 challenge was won by Gessert *et al.* [9] with a large ensemble of multi-resolution EfficientNets that also used patient metadata, reaching a balanced accuracy of about 0.63 on the (harder, out-of-distribution) official test set. Kassani and Kassani [10] compared VGG, ResNet, Xception and DenseNet [11] for melanoma detection

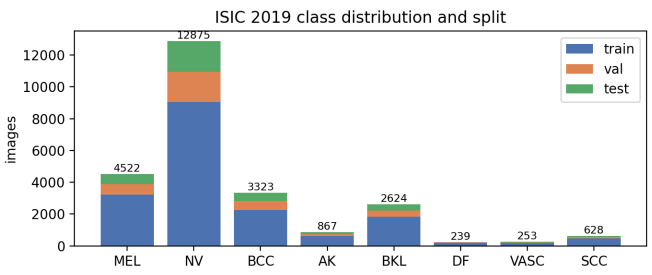


Fig. 1. Class distribution of ISIC 2019 and the lesion-grouped train/validation/test split.

and found ResNet-50 the most accurate of the classical backbones. ImageNet pre-training [12] is used almost universally; Raghu *et al.* [13] showed that, for medical images, its main benefit is faster convergence rather than a large accuracy gain, and that lightweight networks can match large ones.

Transformers in medical imaging. The ViT [5] applies the self-attention of [14] to image patches; Swin [6] restores locality and a feature pyramid through shifted windows, which makes it a drop-in backbone for dense tasks (e.g. TransUNet [15]). Matsoukas *et al.* [7] found that, on small medical datasets, ViTs match CNNs only when ImageNet-pretrained. Aladhadh *et al.* [16] report competitive results of a ViT on HAM10000. Most of these studies compare a single metric and rarely report the cost side; our study fills that gap on ISIC 2019 with a strictly shared protocol.

III. Methodology

A. Dataset and split

We use the public ISIC 2019 training set (25,331 images, eight classes), the only part with public labels. The official `lesion_id` metadata is used to build a *lesion-grouped*, class-stratified split of 17,727/3,844/3,760 images (train/validation/test, ≈70/15/15%), so that no lesion appears in two splits (Fig. 1). The split is fixed (seed 42) and shared by all models.

B. Pre-processing and augmentation

Images are resized once to a shorter side of 256 px and centre-cropped to 256×256. During training we apply random resized crops to 224×224 (scale 0.7–1.0), horizontal and vertical flips, rotations up to 20° and mild colour jitter; at evaluation the centre 224×224 crop is used. Pixels are normalised with ImageNet statistics. The same pipeline feeds every model.

C. Models

All backbones are taken from `timm` [17] with ImageNet-1k weights and a new eight-way linear head, and are fine-tuned end to end:

- **ResNet-50** [18]: the reference residual CNN (23.5 M parameters, 8.17 GFLOPs).
- **EfficientNet-B0** [19]: a compound-scaled, mobile-oriented CNN (4.0 M, 0.77 GFLOPs).
- **Swin-T** [6]: a hierarchical vision transformer with shifted-window attention (27.5 M, 8.98 GFLOPs), chosen over ViT-B because its cost is comparable to ResNet-50, which keeps the comparison fair.

As an exploratory variant we also evaluate a **soft-voting ensemble** that averages the softmax outputs of the three fine-tuned models.

D. Training protocol (identical for all models)

Batch size 32, AdamW [20] with learning rate 10^{-4} and weight decay 0.05, one warm-up epoch followed by cosine decay, at most 12 epochs with early stopping (patience 4) on validation macro-F1, gradient clipping at 1.0 and mixed-precision training [21]. To counter class imbalance the cross-entropy loss is weighted by the inverse square root of the class frequency and smoothed with $\epsilon = 0.1$ [22]. The checkpoint with the best validation macro-F1 is evaluated once on the test split. All runs use a single NVIDIA RTX 4070 Laptop GPU (8 GB).

E. Metrics

Accuracy and *macro-F1* (required by the course), *balanced accuracy* (the official ISC 2019 metric), macro one-vs-rest *AUC* and per-class F1 on the test split. *Cost*: trainable parameters, GFLOPs per image, peak training VRAM, mean epoch time, total training time, the epoch of the best checkpoint and the number of epochs needed to reach 95% of the best validation F1 (convergence speed), and fp16 inference latency at batch 1 and throughput at batch 32. *Interpretability*: Grad-CAM [23] on the last convolutional stage of the CNNs and on the last Swin block (tokens reshaped to a 7×7 map).

IV. Results

A. Overall performance and cost

Table I summarises the test results. Training curves are shown in Fig. 2 and the accuracy/cost trade-off in Fig. 3.

B. Per-class behaviour

Table II and Fig. 4 report the per-class F1, and Fig. 5 the confusion matrices.

C. Interpretability

Fig. 6 shows Grad-CAM maps for one test image per class.

V. Discussion

A. Which model is "better"?

On the decision metrics the ranking is unambiguous: Swin-T is the best single backbone (76.1% accuracy, 63.9% balanced accuracy, 64.1% macro-F1), followed by EfficientNet-B0 (57.8%) and ResNet-50 (55.1%). The gap between the transformer and the best CNN is 6.3 macro-F1 points, which is larger than the 2.7 points separating the two CNNs, so on this dataset the family label does matter—but not in the direction that is often assumed for small medical datasets [7]. Swin-T also converges faster (64.1% macro-F1 is already reached at epoch 2 of 12 in validation, Fig. 2), which we attribute to its

TABLE I

TEST-SPLIT RESULTS AND COMPUTATIONAL COST UNDER THE SHARED PROTOCOL (FP16 LATENCY ON AN RTX 4070 LAPTOP; "EP. 95%" = EPOCHS TO REACH 95% OF THE BEST VALIDATION MACRO-F1).

Model	Type	Acc.	Bal. Acc.	Macro-F1	AUC	Params (M)	GFLOPs	Lat. (ms)	img/s	Train (min)	VRAM (MB)	Best ep.	Ep. 95%
ResNet-50	CNN	0.711	0.601	0.551	0.918	23.5	8.17	29.1	724	20.4	1807	12	6
EfficientNet-B0	CNN	0.738	0.610	0.578	0.906	4.0	0.77	36.9	511	13.7	1500	8	5
Swin-T	Transformer	0.761	0.639	0.641	0.900	27.5	8.98	71.2	554	23.4	2667	8	6
Ensemble (soft voting)	Hybrid	0.778	0.680	0.665	0.942	55.1	17.92	137.1	194	–	–	–	–

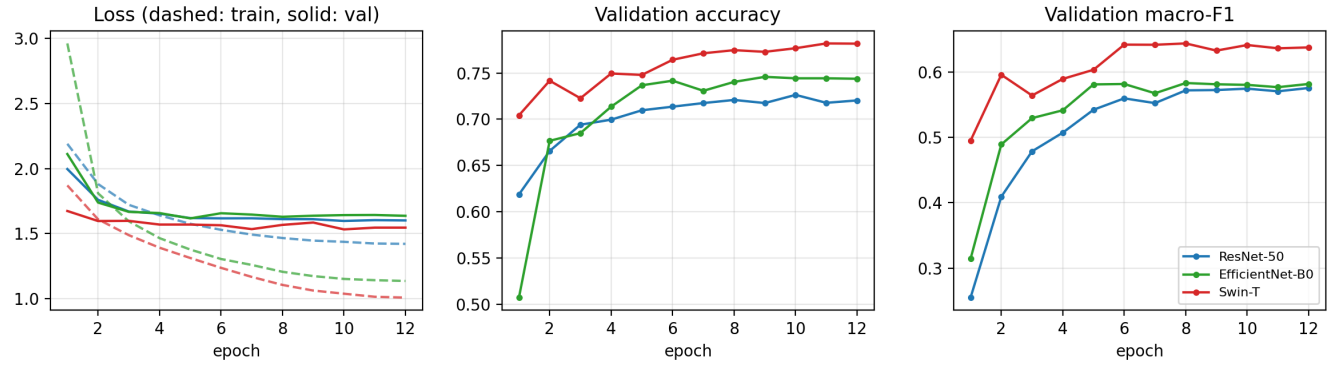


Fig. 2. Training/validation loss, validation accuracy and validation macro-F1 per epoch.

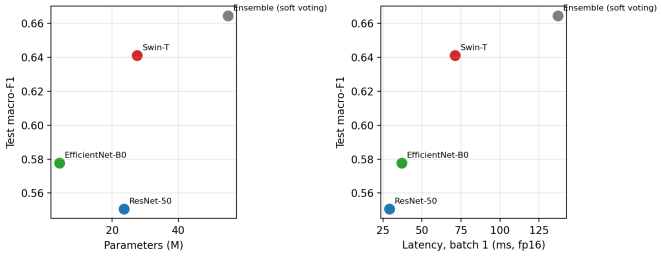


Fig. 3. Test macro-F1 versus parameters (left) and batch-1 latency (right).

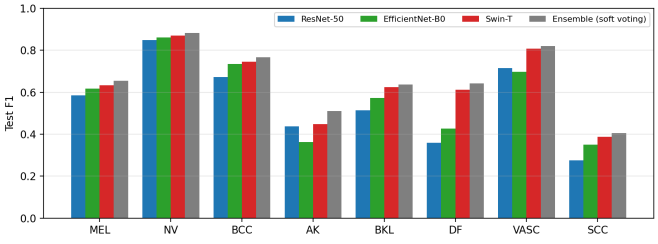


Fig. 4. Per-class test F1 of each model.

TABLE II

PER-CLASS TEST F1.

Model	MEL	NV	BCC	AK	BKL	DF	VASC	SCC
ResNet-50	0.58	0.85	0.67	0.44	0.51	0.36	0.71	0.28
EfficientNet-B0	0.62	0.86	0.73	0.36	0.57	0.43	0.70	0.35
Swin-T	0.63	0.87	0.75	0.45	0.62	0.61	0.81	0.39
Ensemble (soft voting)	0.65	0.88	0.77	0.51	0.64	0.64	0.82	0.40

stronger ImageNet features and to the hierarchical windowed attention, which lets it integrate colour and texture cues over the whole lesion rather than within a fixed receptive field.

The ranking by macro AUC is the *opposite* (ResNet-50 0.918 > EfficientNet-B0 0.906 > Swin-T 0.900). AUC measures the quality of the probability ranking of each class against the rest, whereas accuracy and F1 measure the hard arg-max decision. Swin-T fits the training set much more tightly (its training loss ends at 1.00 versus 1.42 for ResNet-50 with the same label smoothing), which produces sharper, more confident posteriors: better decisions on the hard examples but slightly worse calibrated rankings on the easy ones. This is a reminder that a single number can hide a trade-off, and that a clinical triage tool that thresholds probabilities should be evaluated with AUC and calibration, not only with accuracy.

B. *Cost: FLOPs are not latency*

EfficientNet-B0 has 6× fewer parameters and 10× fewer FLOPs than ResNet-50, trains in 14 instead of 20 minutes and uses the least VRAM (1.5 GB). Yet on the GPU it is *slower* at inference (36.9 ms versus 29.1 ms at batch 1; 511 versus 724 images/s at batch 32). Its depthwise-separable convolutions are memory-bound and launch many small kernels that cannot exploit tensor cores, so the theoretical savings do not materialise on this hardware; they would on a CPU or mobile NPU, which is the setting EfficientNet was designed for [19]. Swin-T has almost the same FLOPs as ResNet-50 (8.98 versus 8.17 GFLOPs) but is 2.4× slower at batch 1, needs 1.5× more VRAM during training (2.6 versus 1.8 GB) and the longest wall-clock time (23 min). The accuracy gain of the transformer therefore has a concrete price: roughly 2.4× the latency and 1.5× the training memory of the residual CNN. In a batch setting (offline screening of a clinic's daily images)

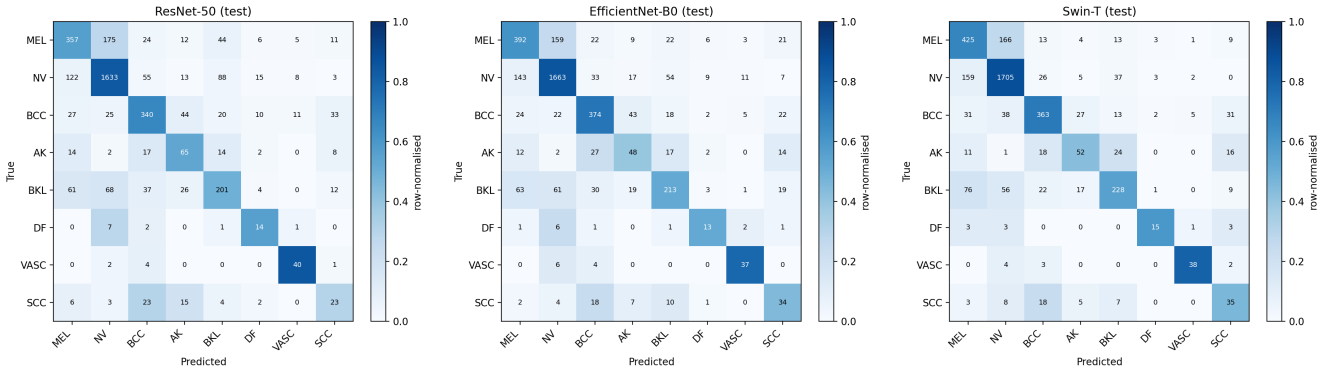


Fig. 5. Row-normalised confusion matrices on the test split (cell labels are raw counts).

this price is negligible; in an interactive or embedded setting it is the deciding factor.

C. Generalisation across classes

All models are strong on the majority class NV (F1 0.85–0.88) and on the visually distinctive VASC and BCC, and weak on the three rare or morphologically overlapping diagnoses AK, SCC and DF (Table II). The confusion matrices (Fig. 5) show two persistent error clusters: MEL↔NV (166 melanomas predicted as nevi and 159 nevi predicted as melanoma by Swin-T) and the keratinocyte group AK↔BKL↔SCC, which is also where dermatologists disagree most. Inverse-square-root class weighting raised the recall of the rare classes (DF recall 0.60, VASC 0.81, SCC 0.46 for Swin-T) without collapsing the majority class, but the balanced accuracy of 0.60–0.64 shows that the imbalance is only partly solved. Swin-T is the most *uniform* model: it has the best F1 in seven of the eight classes and is the only backbone that lifts DF above 0.6, suggesting that its global receptive field helps on classes defined by subtle, lesion-wide texture rather than by a local pattern.

D. What the models look at

The Grad-CAM maps (Fig. 6) are consistent with that reading. The two CNNs produce compact, centred blobs on the pigmented core of the lesion, which is adequate for NV, BCC, DF and VASC but misses the periphery that distinguishes MEL from NV and AK from BKL. Swin-T produces sparser, multi-focal maps (e.g. two separated hot spots on the melanoma, a ring on the vascular lesion) that follow the irregular border and the asymmetric pigment distribution—precisely the "ABCD" cues used by clinicians. The failures are informative too: on the BKL example all three models fire on the darkest region and predict melanoma, and on the SCC example EfficientNet-B0 spreads its evidence over the background, where hair and scale artefacts are present. Such artefact-driven activations are a known risk on ISIC [8] and argue for lesion segmentation or artefact removal in a deployed system.

E. Are the families complementary?

The soft-voting ensemble improves on the best single model by 2.3 macro-F1 points (66.5% versus 64.1%), by 4.0 balanced-accuracy points and raises the macro AUC to 0.942, the best value in the study. Among the pairwise ensembles, the two CNNs together reach only 61.5% macro-F1, whereas EfficientNet-B0+Swin-T reaches 66.4%, almost the value of the full ensemble; i.e. nearly all the gain comes from combining a CNN with the transformer. The two families make different mistakes, which is the strongest practical argument for hybrid systems, at the cost of 137.1 ms per image.

F. Comparison with the literature and limitations

Our balanced accuracies (0.60–0.68) are in the range reported for single ImageNet-pretrained models on ISIC 2019 at 224 px, and below the 0.63–0.74 obtained by the challenge winner with multi-resolution EfficientNet ensembles trained on far more compute and with metadata [9]; note that our numbers are measured on an internal, in-distribution split and are therefore not directly comparable to the official out-of-distribution test set. Limitations: a single seed per model (differences below $\approx$1 point should not be over-interpreted), a fixed 224 px resolution that discards fine dermoscopic structure, no use of age/sex/site metadata, no test-time augmentation, and a learning rate that was not tuned per architecture (a per-model search would most likely narrow, not widen, the gaps).

VI. Conclusions and Future Work

We compared a residual CNN, an efficiency-oriented CNN and a hierarchical vision transformer on ISIC 2019 under a strictly shared, lesion-grouped protocol and measured accuracy, cost and interpretability side by side. Three conclusions follow. (i) With ImageNet pre-training, the transformer is the most accurate and the most class-uniform backbone on 25k dermoscopic images (64.1% macro-F1, 63.9% balanced accuracy), contradicting the view that transformers need very large medical datasets; its evidence maps also align better with clinical ABCD cues. (ii) That accuracy costs about 2.4$\times$ the inference latency and 1.5$\times$ the training memory of ResNet-50, and the cheapest model by FLOPs (EfficientNet-B0) is not the fastest on a GPU, so deployment decisions

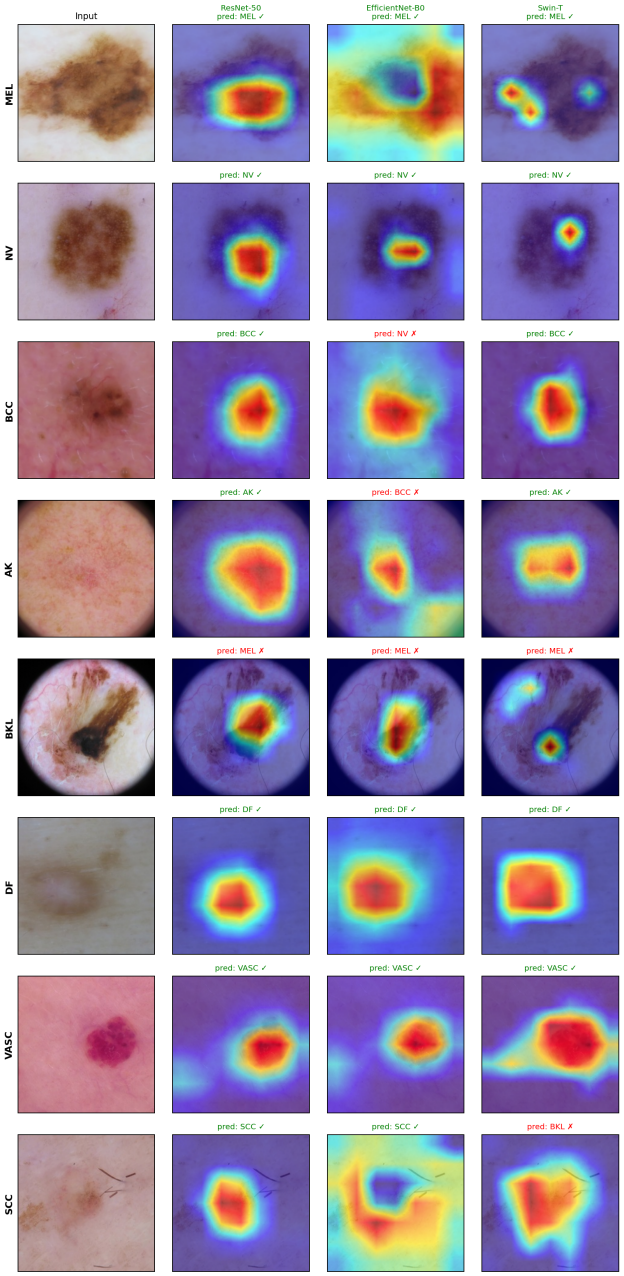


Fig. 6. Grad-CAM evidence maps (target = true class) for one test image per class. Green/red titles mark correct/incorrect predictions.

must be based on measured latency on the target hardware rather than on parameter counts. (iii) CNNs and transformers make complementary errors: a simple soft-voting ensemble gives the best macro-F1 (66.5%) and AUC (0.942) of the study. Our practical recommendation is EfficientNet-B0 for CPU/edge budgets, Swin-T when a GPU is available, and the CNN+transformer ensemble for offline screening where accuracy dominates.

Future work should (a) repeat the study with several seeds and at 384–512 px, where fine dermoscopic structures become visible; (b) add patient metadata and test-time augmentation, both of which helped the challenge winners [9]; (c) evaluate calibration and decision thresholds for triage, given the AUC/F1 disagreement observed here; (d) replace ImageNet with dermoscopy-specific self-supervised pre-training (DINO/MAE on unlabelled ISIC archives), which targets the domain gap directly; and (e) validate externally on the official ISIC 2019 test set and on clinical (non-dermoscopic) datasets such as PAD-UFES-20 to measure true generalisation.

Reproducibility

Code, split file, training logs and all figures are available in the project repository (`src/`, `results/`); `run_all.ps1` regenerates every number in this paper from the public ISIC 2019 download.

References

[1] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.

[2] N. C. F. Codella, D. Gutman, M. E. Celebi, B. Helba, M. A. Marchetti, S. W. Dusza, A. Kalloo, K. Liopyris, N. Mishra, H. Kittler, and A. Halpern, "Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (ISBI), hosted by the international skin imaging collaboration (ISIC)," in *Proc. IEEE Int. Symp. Biomedical Imaging (ISBI)*, 2018, pp. 168–172.

[3] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Scientific Data*, vol. 5, p. 180161, 2018.

[4] M. Combalia, N. C. F. Codella, V. Rotemberg, B. Helba, V. Vilaplana, O. Reiter, C. Carrera, A. Barreiro, A. C. Halpern, S. Puig, and J. Malvehy, "BCN20000: Dermoscopic lesions in the wild," *arXiv preprint arXiv:1908.02288*, 2019.

[5] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[6] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 10 012–10 022.

[7] C. Matsoukas, J. F. Haslum, M. Söderberg, and K. Smith, "Is it time to replace CNNs with transformers for medical images?" *arXiv preprint arXiv:2108.09038*, 2021.

[8] B. Cassidy, C. Kendrick, A. Brodzicki, J. Jaworek-Korjakowska, and M. H. Yap, "Analysis of the ISIC image datasets: Usage, benchmarks and recommendations," *Medical Image Analysis*, vol. 75, p. 102305, 2022.

[9] N. Gessert, M. Nielsen, M. Shaikh, R. Werner, and A. Schlaefer, "Skin lesion classification using ensembles of multi-resolution EfficientNets with meta data," *MethodsX*, vol. 7, p. 100864, 2020.

[10] S. H. Kassani and P. H. Kassani, "A comparative study of deep learning architectures on melanoma detection," *Tissue and Cell*, vol. 58, pp. 76–83, 2019.

[11] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708.

[12] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.

[13] M. Raghu, C. Zhang, J. Kleinberg, and S. Bengio, "Transfusion: Understanding transfer learning for medical imaging," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.

[14] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[15] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, and Y. Zhou, "TransUNet: Transformers make strong encoders for medical image segmentation," *arXiv preprint arXiv:2102.04306*, 2021.

[16] S. Aladhadh, M. Alsadhan, M. Alghamdi, M. Alsaedi, M. Alshammari, and T. Ahmad, "An effective skin cancer classification mechanism via medical vision transformer," *Sensors*, vol. 22, no. 11, p. 4008, 2022.
[17] R. Wightman, "PyTorch image models," https://github.com/huggingface/pytorch-image-models, 2019.
[18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
[19] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.
[20] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.
[21] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, and H. Wu, "Mixed precision training," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2018.
[22] R. Müller, S. Kornblith, and G. E. Hinton, "When does label smoothing help?" in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
[23] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 618–626.